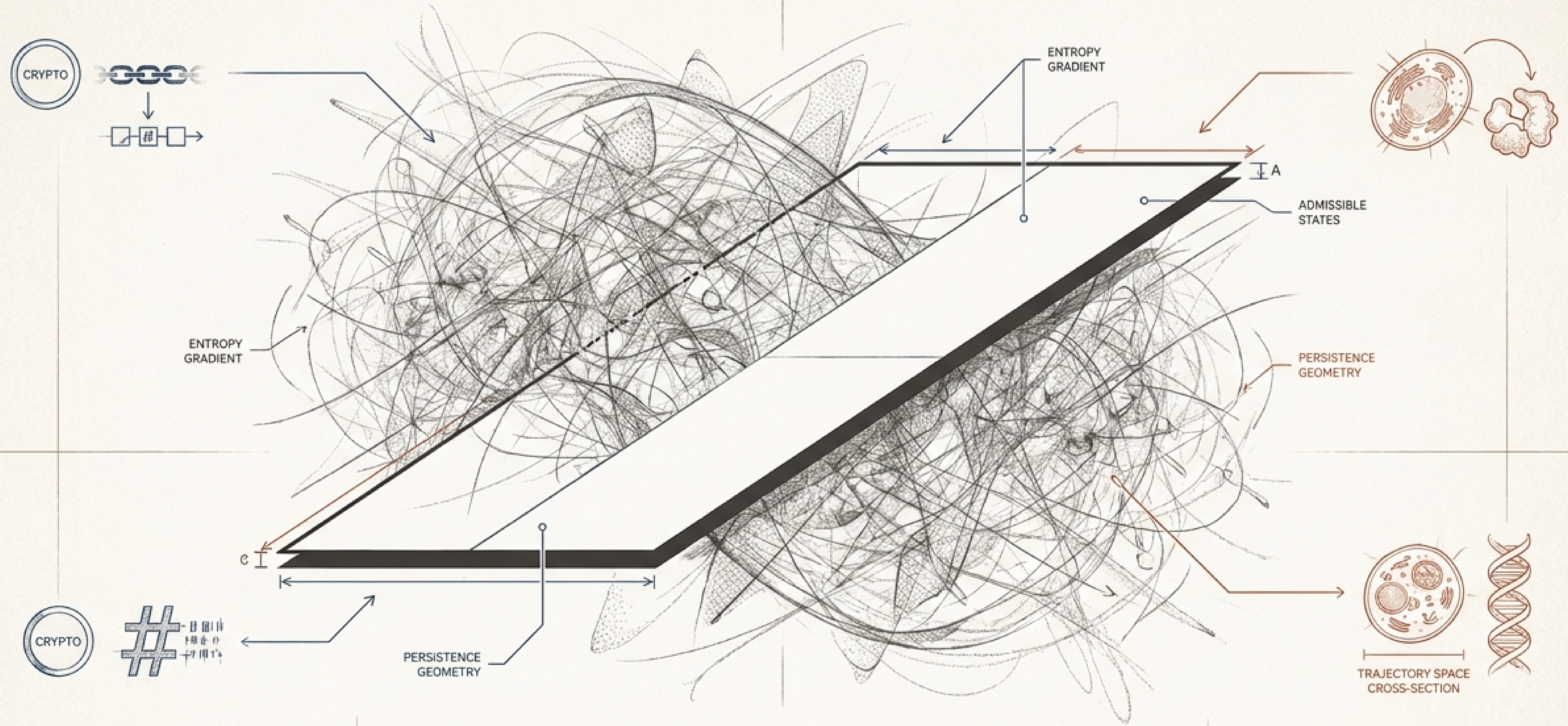


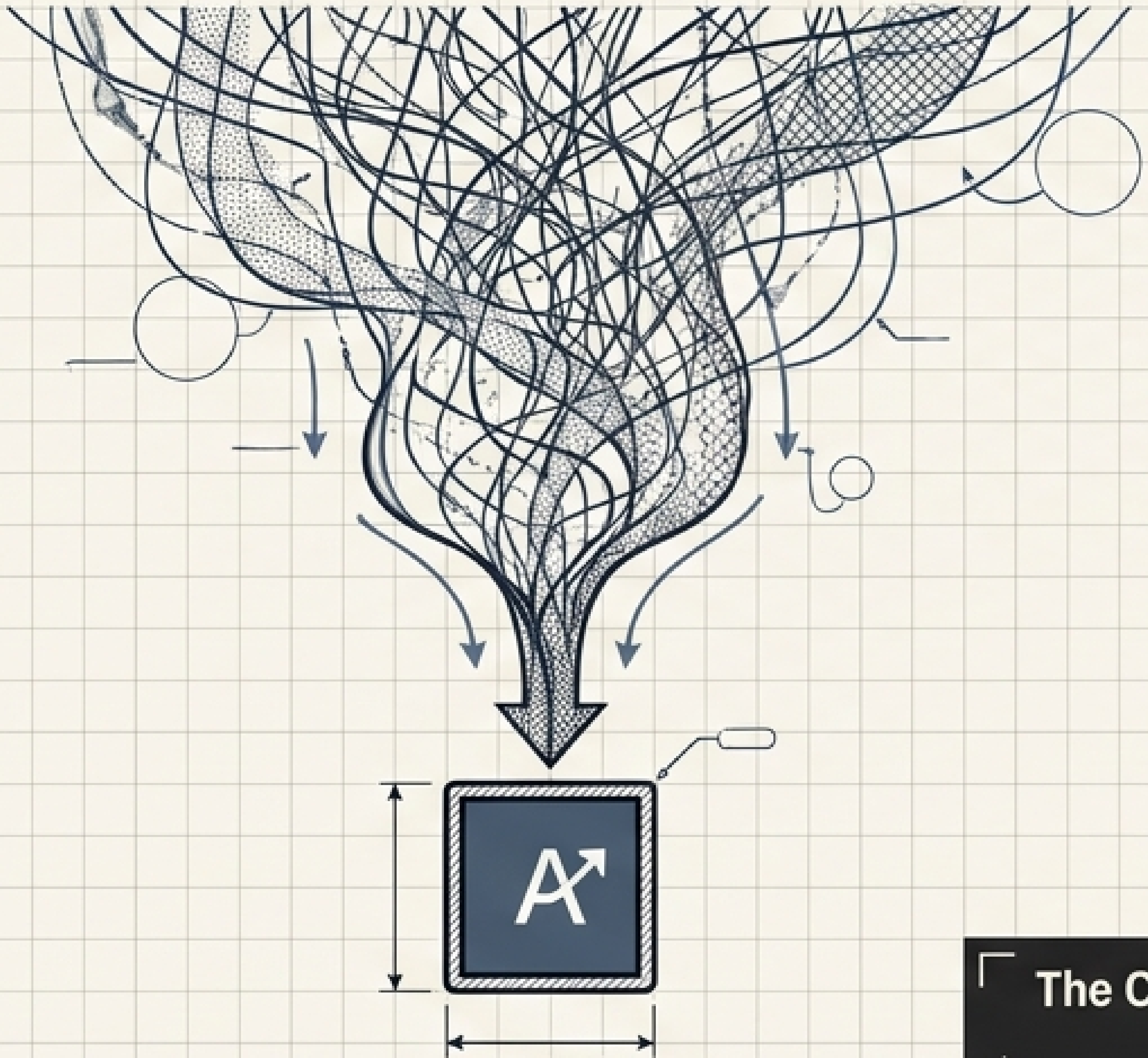
Function Survives Collapse

Admissibility, Entropy Flow, and the Geometry of Persistent Computation

A unified theory of cryptography and molecular biology.

Cryptography (2025)

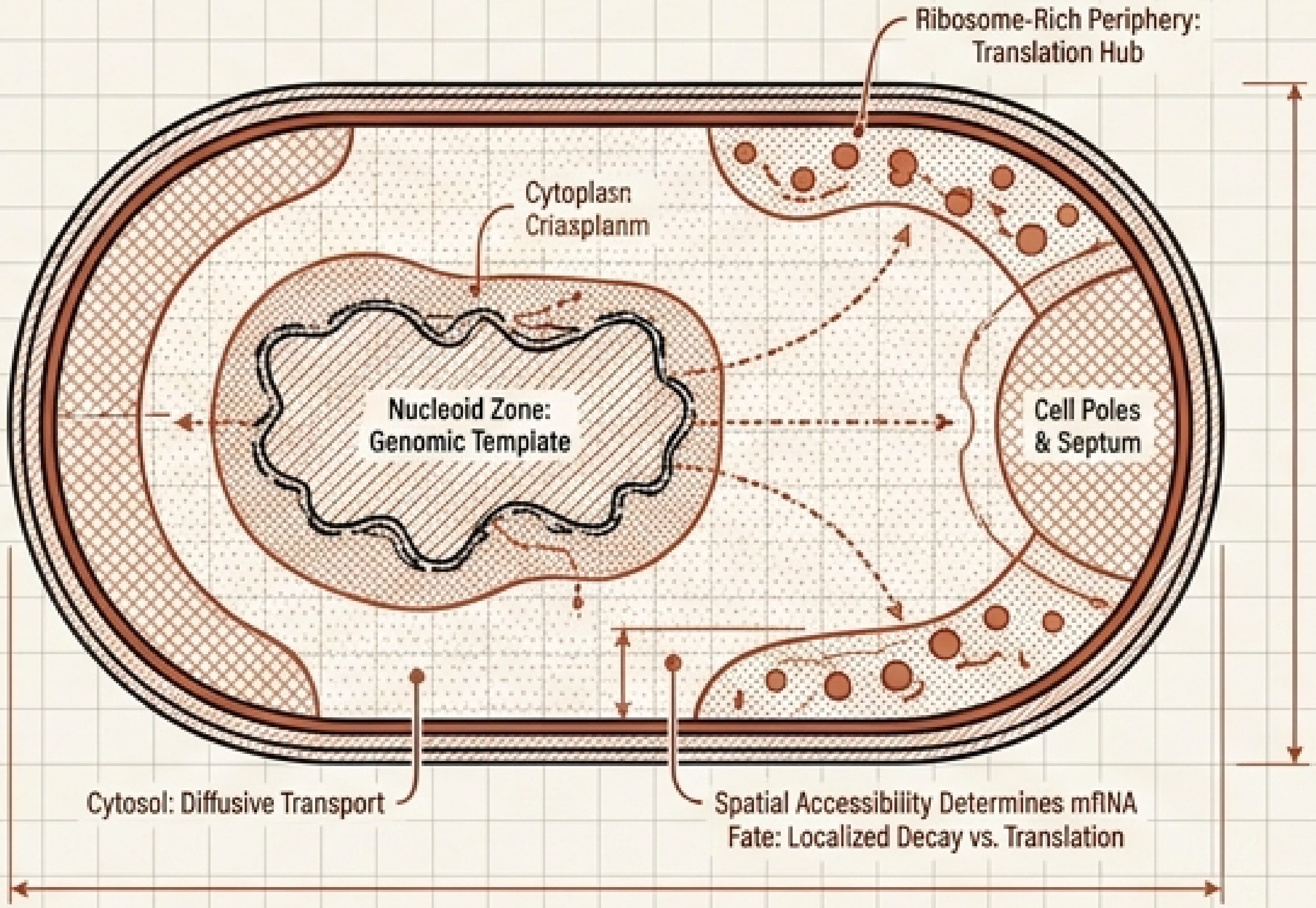
Bitansky et al.: Arbitrary secure computation is achieved through anonymous additive aggregation. Local provenance is irreversibly destroyed, yet computable functional invariants persist.



Immutable Functional Invariant
(Aggregated Output)

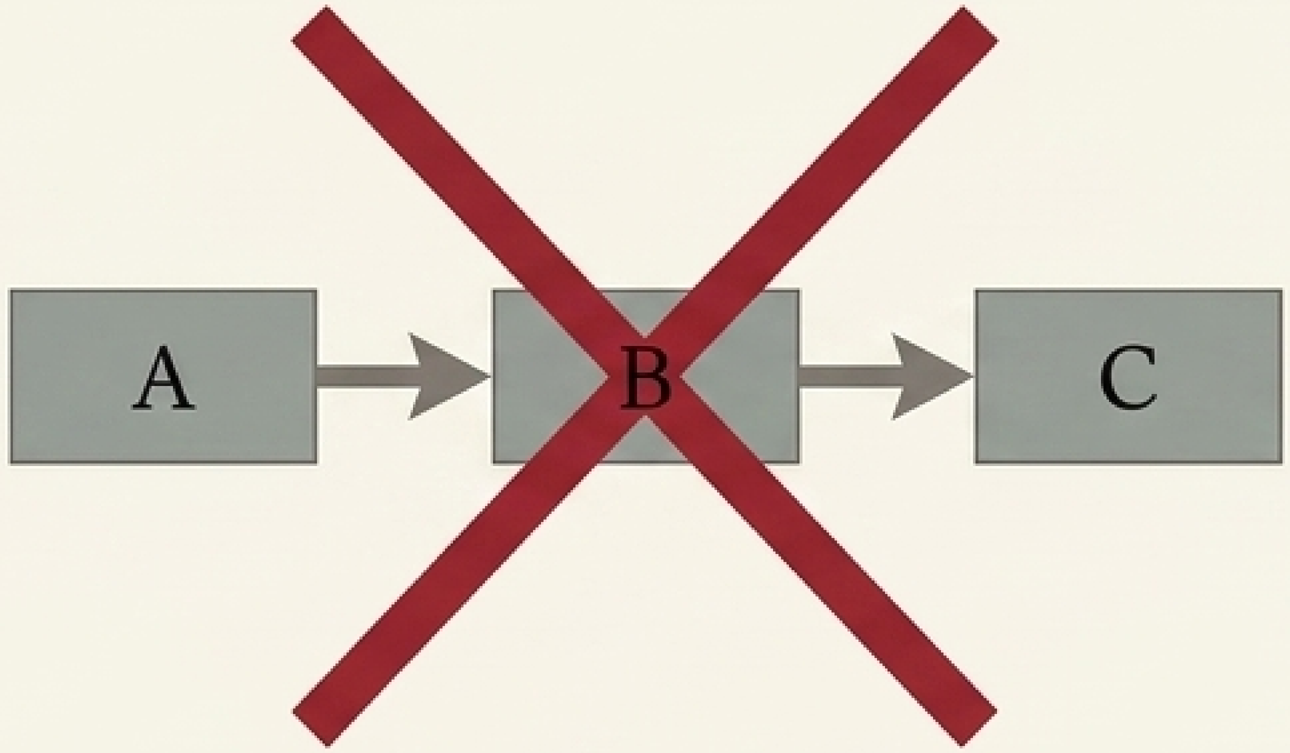
Molecular Biology (2026)

Kim et al.: Gene expression is not a linear pipeline but a geometry-constrained coordination of decoupled flows. mRNA fate depends critically on spatial accessibility.



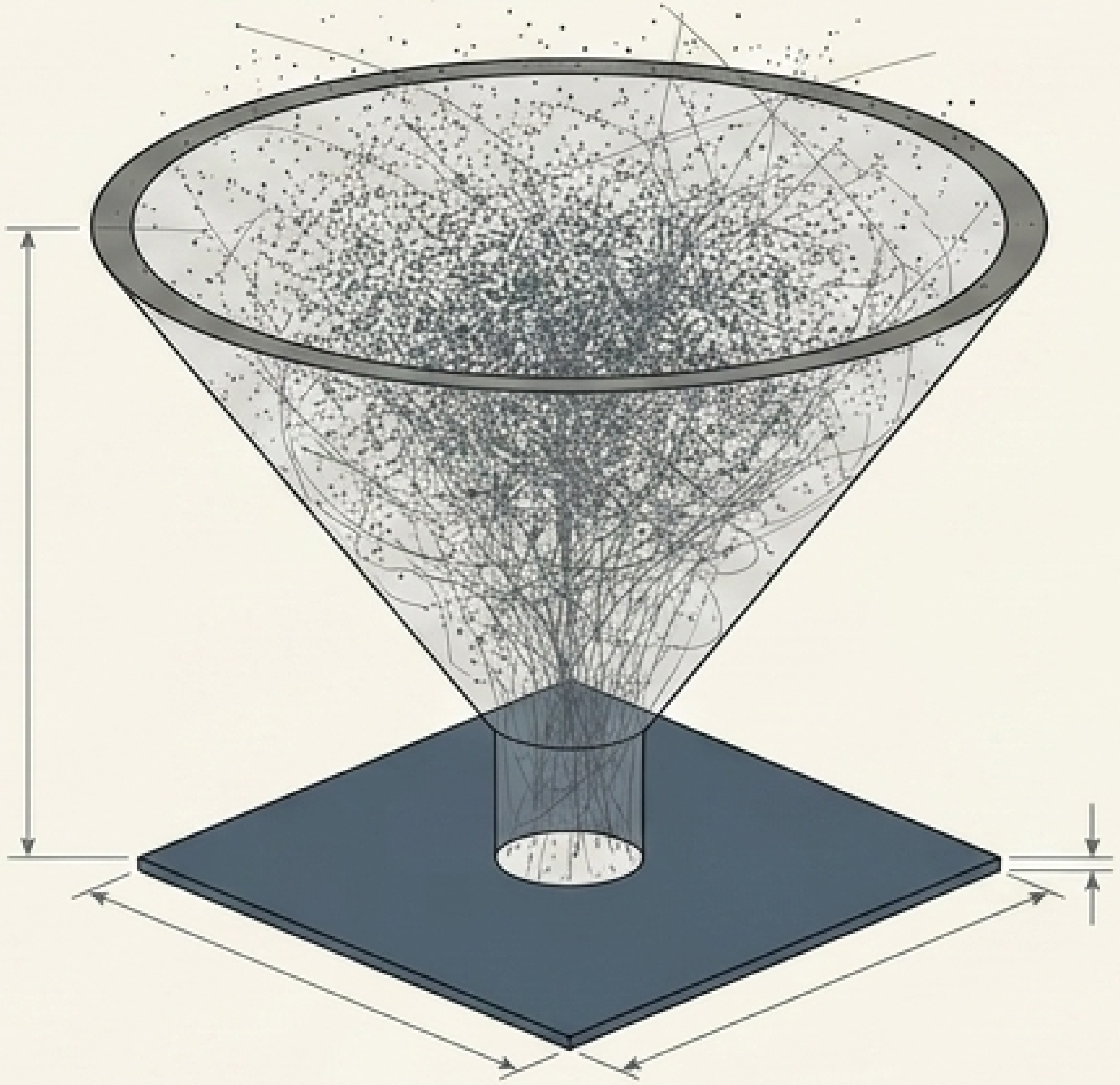
The Convergent Phenomenon: In both systems, functional output is infinitely more stable than the microscopic paths that created it.

THE PIPELINE ONTOLOGY



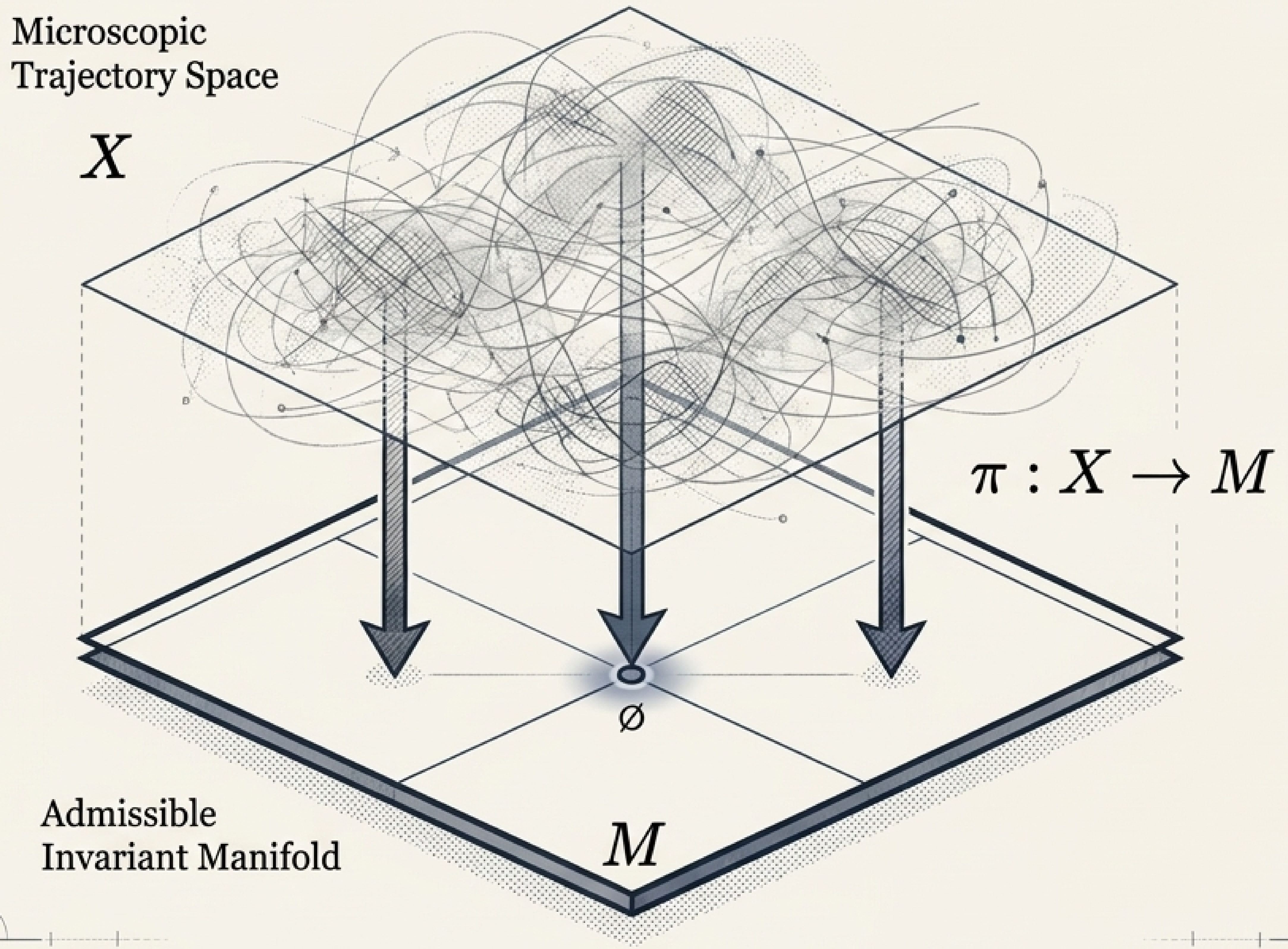
The classical assumption: Secure channels in cryptography; Central Dogma ($DNA \rightarrow RNA \rightarrow$ Protein) in biology.

THE PROJECTION ONTOLOGY



The new framework: Massive microscopic chaos collapsing into a mathematically stable functional output.

The Reorientation: Stop asking how information travels from source to output. Ask which functionals of the source are preserved when microscopic recoverability collapses.



1. Functional Preservation:

$$f_\alpha(\pi(x)) = f_\alpha(x).$$

The system discards everything except what is needed for the designated output.

2. Irreversibility:

Multiple distinct trajectories map to the exact same invariant on the manifold.

3. Entropy Non-Decrease:

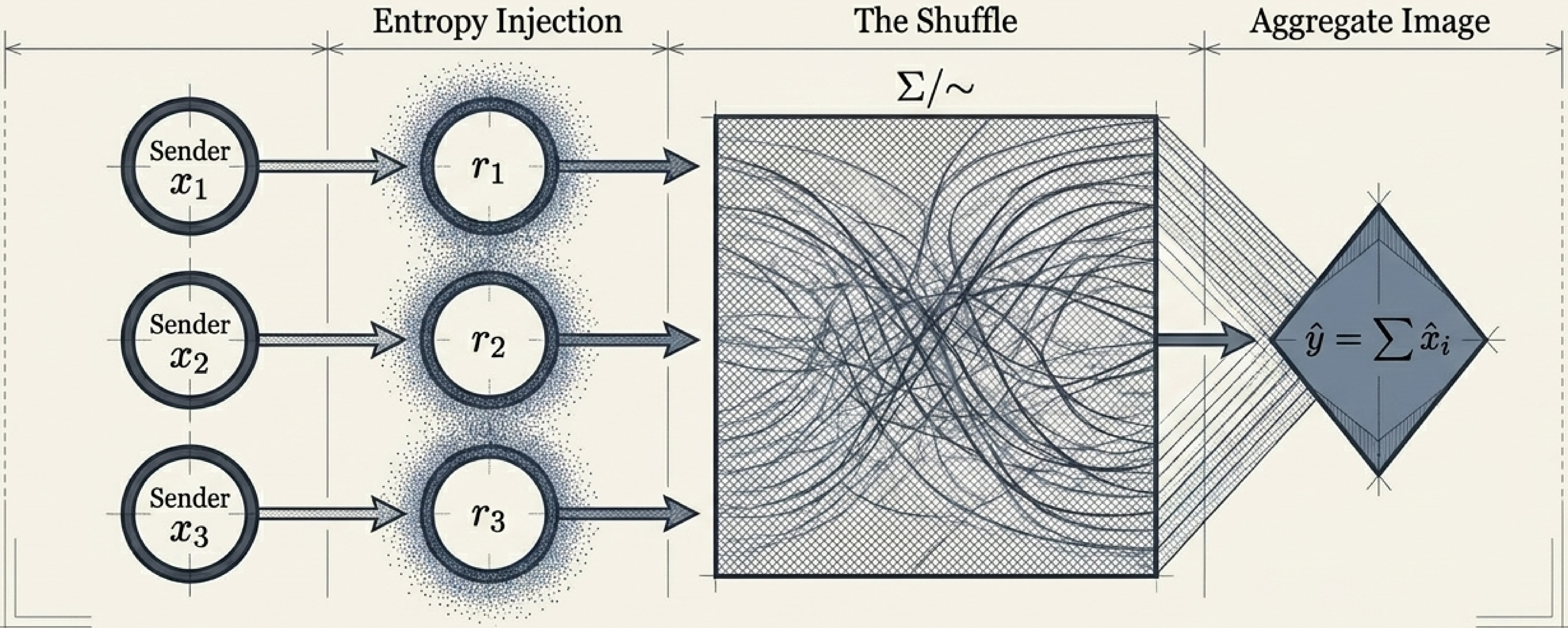
$$H(\pi(X)) \leq H(X).$$

The collapse treats the path as fuel and the functional output as the signal.

4. Provenance Destruction:

There is no efficient mathematical or geometric procedure to reverse the flow and recover x from $\pi(x)$.

The Convergent Phenomenon: In both systems, functional output is infinitely more stable than the microscopic paths that created it.



The ARE Framework (Halevi et al., 2023)

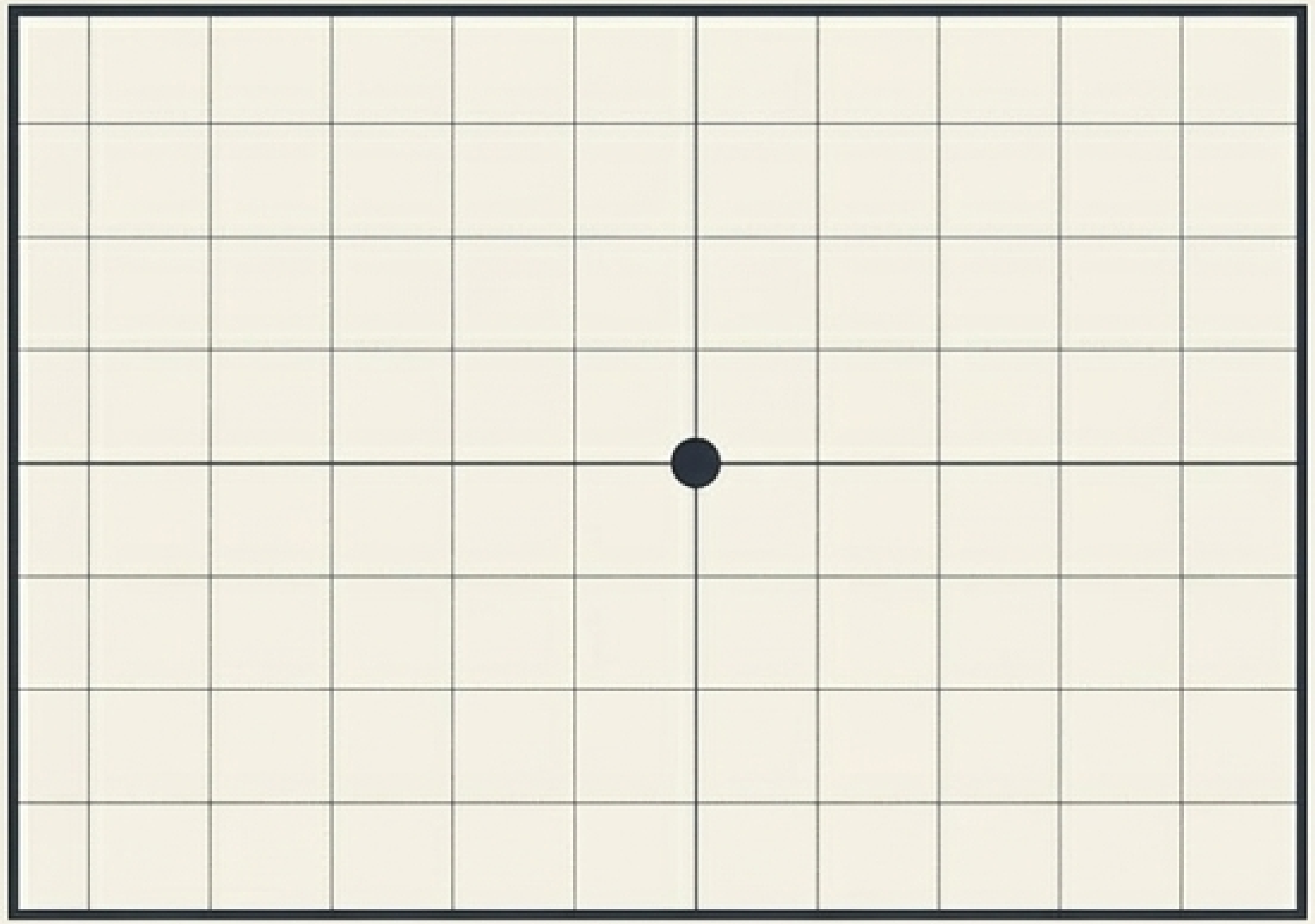
Computation requires no rounds, no authenticated channels, no correlated setup. The only global operation is addition.

The Mechanism

The encoding map $x_i \mapsto \hat{x}_i$ is a local section of the admissible projection. The fibers $\pi^{-1}(\hat{y})$ are arbitrarily large, mathematically forbidding the reconstruction of individual inputs.

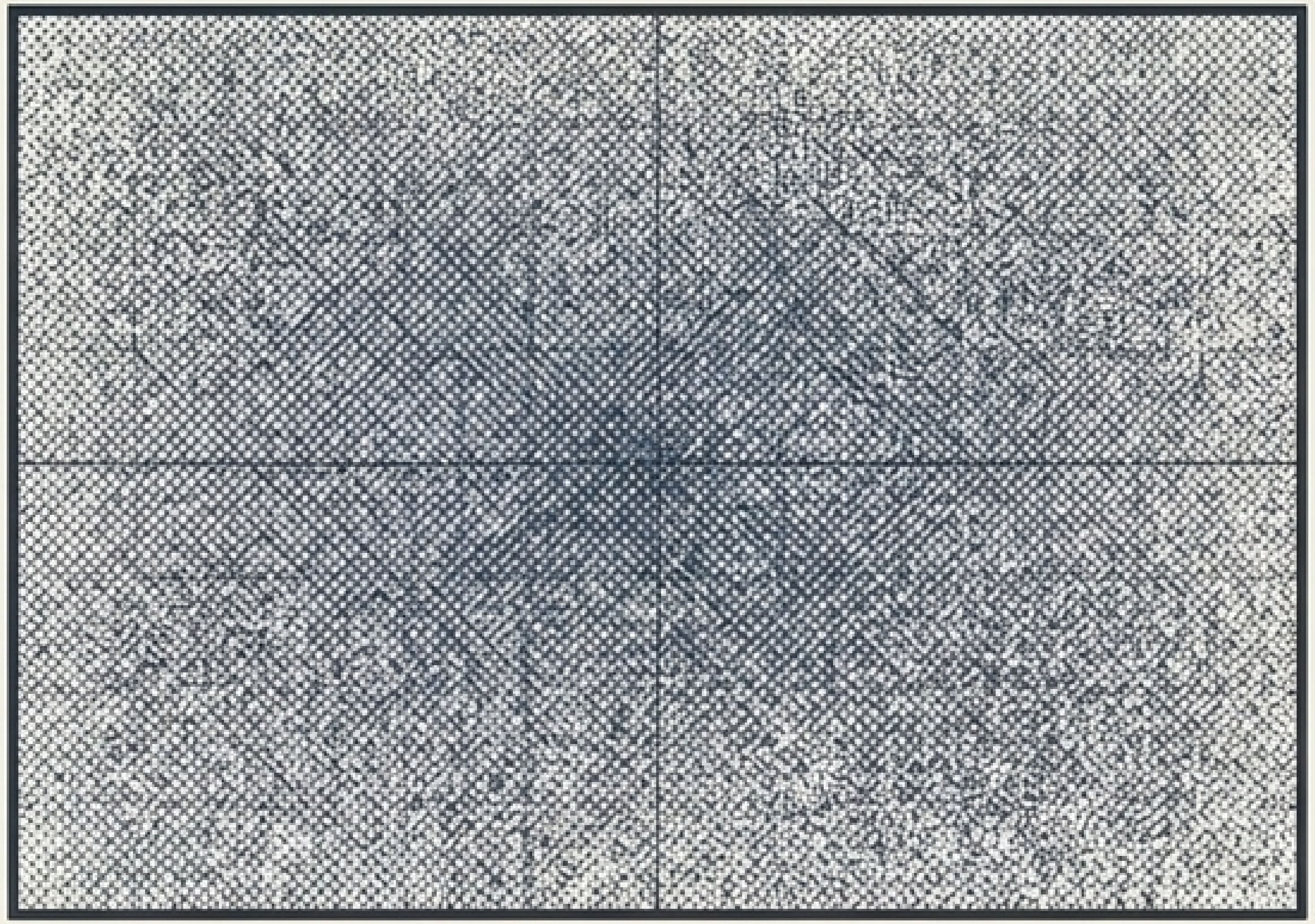
Below Threshold: 0 Active Parties

Identity Element ($0 \in G$)



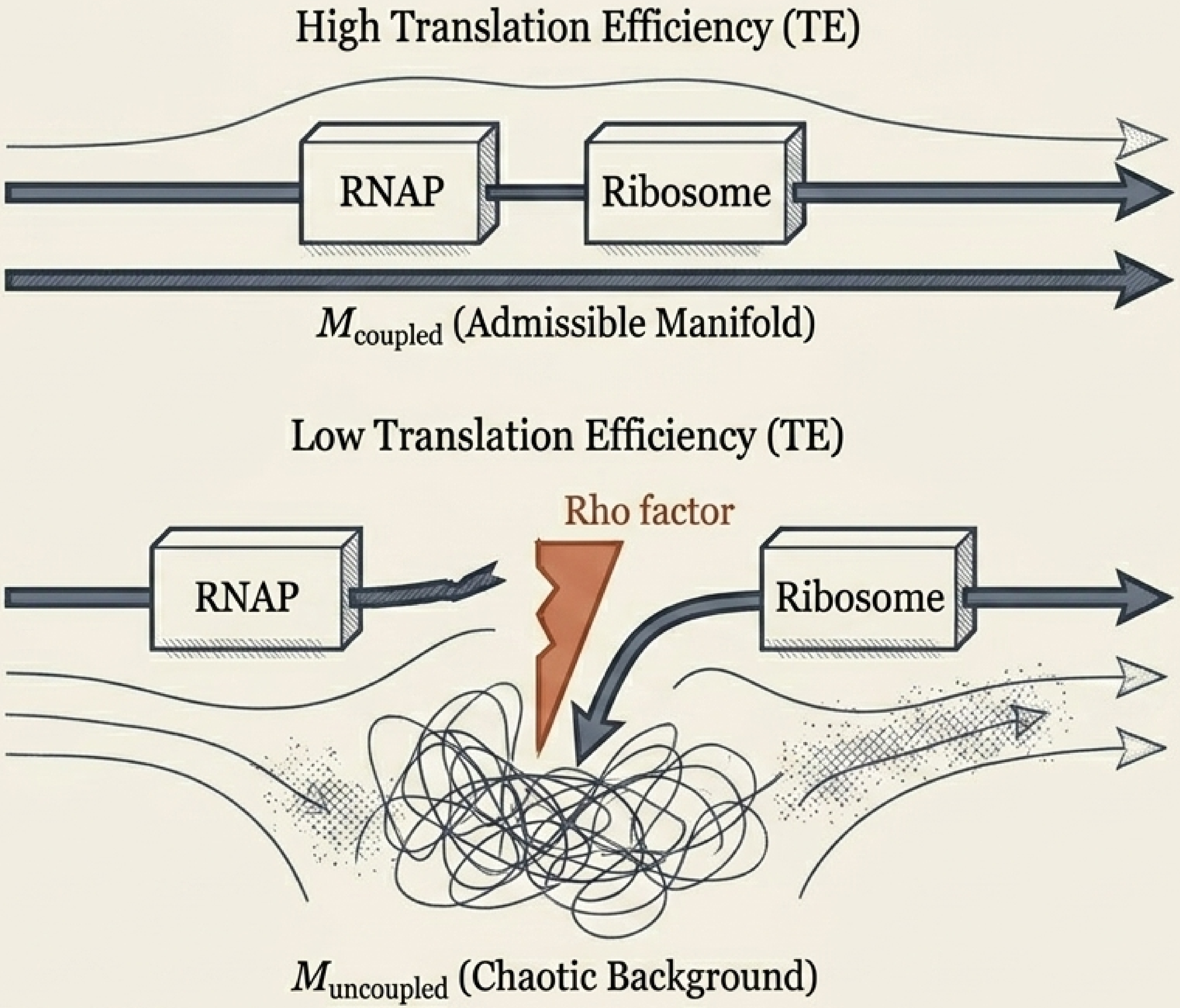
Above Threshold: 1+ Active Parties

Uniform Random Distribution



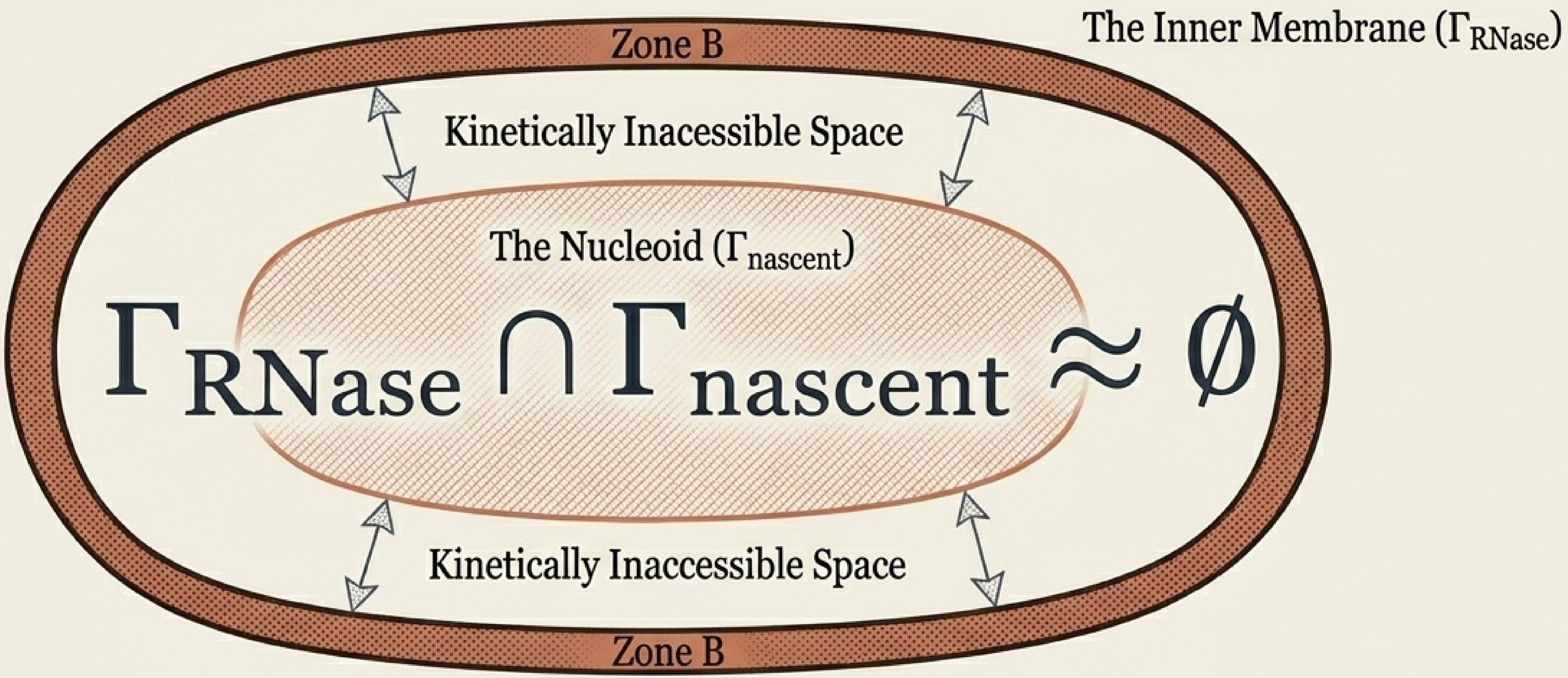
Universality via Entropy Amplification

Bitansky et al. (2025) proved statistical ARE exists for all functions. The phase transition demonstrates that the presence of just one active participant thermalizes the entire sum. A single unit of entropy injection is mathematically sufficient to mask the entire microscopic configuration of the group.



The End of the Central Dogma Pipeline

Kim et al. (2026) show that universal coupling is a myth. Coupling is a field parameter governed by Translation Efficiency. Low TE allows RNAP to outpace the ribosome, driving the trajectory off the manifold. Premature termination probability and degradation kinetics are perfectly correctly correlated ($r = -0.93$).



Membrane Localization as Topological Obstruction

Transcripts survive not because of a protective biochemical shield, but because the cell's spatial geometry strictly forbids the encounter.

The Experimental Proof

When RNase E is experimentally relocated to the cytoplasm, the spatial constraint collapses and co-transcriptional degradation skyrockets sevenfold (from $0.042 \rightarrow 0.31\ \text{min}^{-1}$).

Species-Specific Admissibility Topologies

	Species: *E. coli* / *B. subtilis*	Species: *Caulobacter crescentus*
Primary Ribonuclease Location	Anchored to Inner Membrane	Free in Cytoplasm
Spatial Admissibility Constraint	$\Gamma_{\mathrm{RNase}} \cap \Gamma_{\mathrm{nascent}} \approx \emptyset$	Constraint Collapsed (Substantial Overlap)
Co-Transcriptional Degradation Rate	Negligible ($k_{\mathrm{d1}} \approx 0.042\ \mathrm{min}^{-1}$)	Rapid ($k_{\mathrm{d1}} \approx 1.3\ \mathrm{min}^{-1}$)

Key Insight: Evolution doesn't just find different biochemical solutions; it computes identical functional invariants using entirely distinct geometric accessibility structures.

The Duality of Topological Privacy

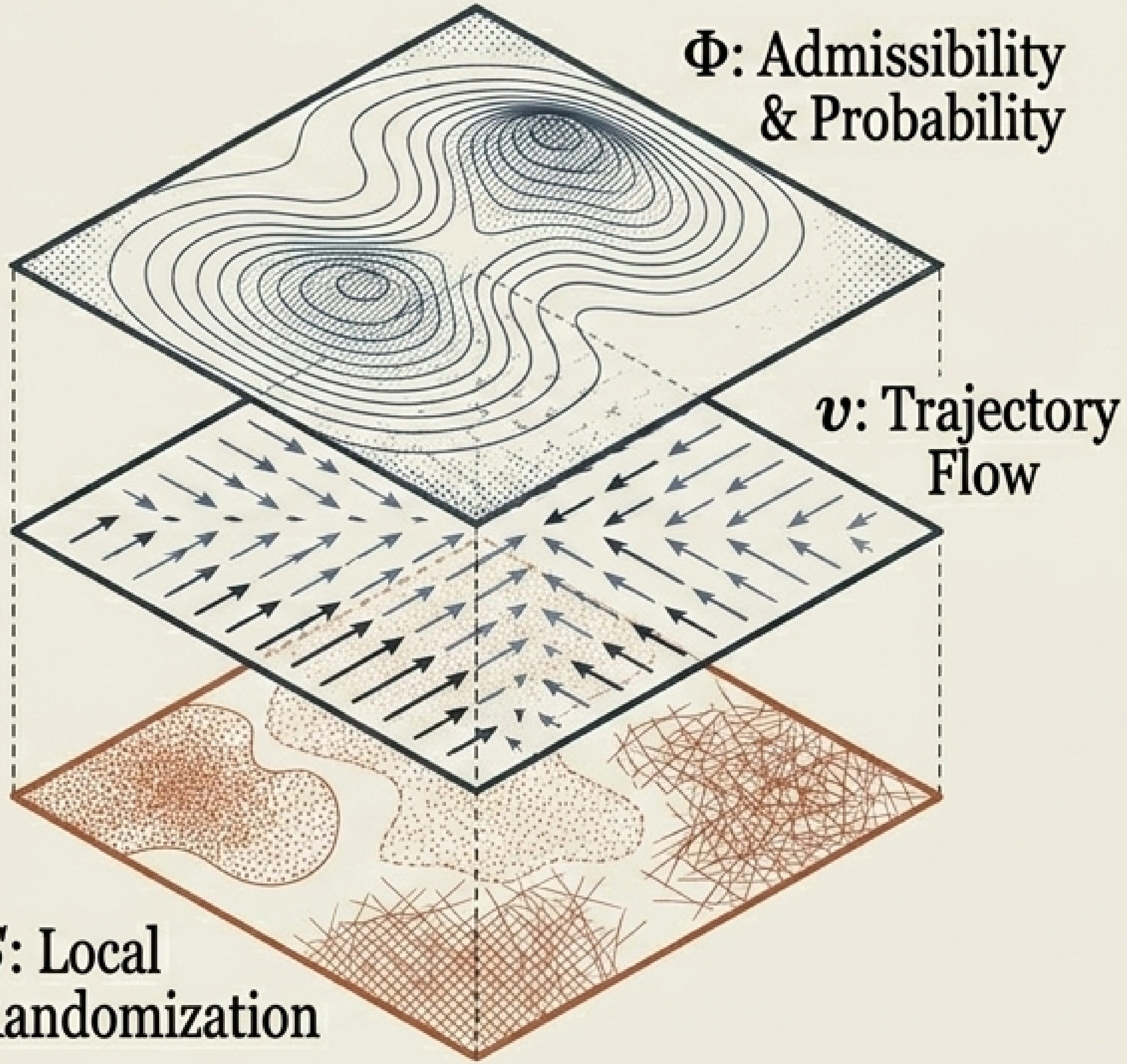
	Additive Randomized Encodings	Bacterial Gene Expression
Mechanism of Privacy	Mixing (Aggregating inputs)	Separation (Geometric spacing)
Entropy Flow	Injection of large fibers & noise	Sharp spatial gradients & restriction
Obstruction Geometry	Cohomological Trivialization (Too many valid paths exist)	Cohomological Obstruction (No valid paths exist)
How Provenance is Destroyed	Interior is crowded with indistinguishable alternatives	Interior is entirely empty of reachable paths

Both mechanisms deny the observer access to the microscopic interior. Both allow functional invariants to persist.

The RSVP (Relativistic Scalar-Vector Plenum) Field Triple

RSVP Unified Field Equation

$$\partial_t \Phi + \nabla \cdot (\Phi v) = -\lambda S$$



The invariant manifold M is the dynamic attractor of this constrained flow.

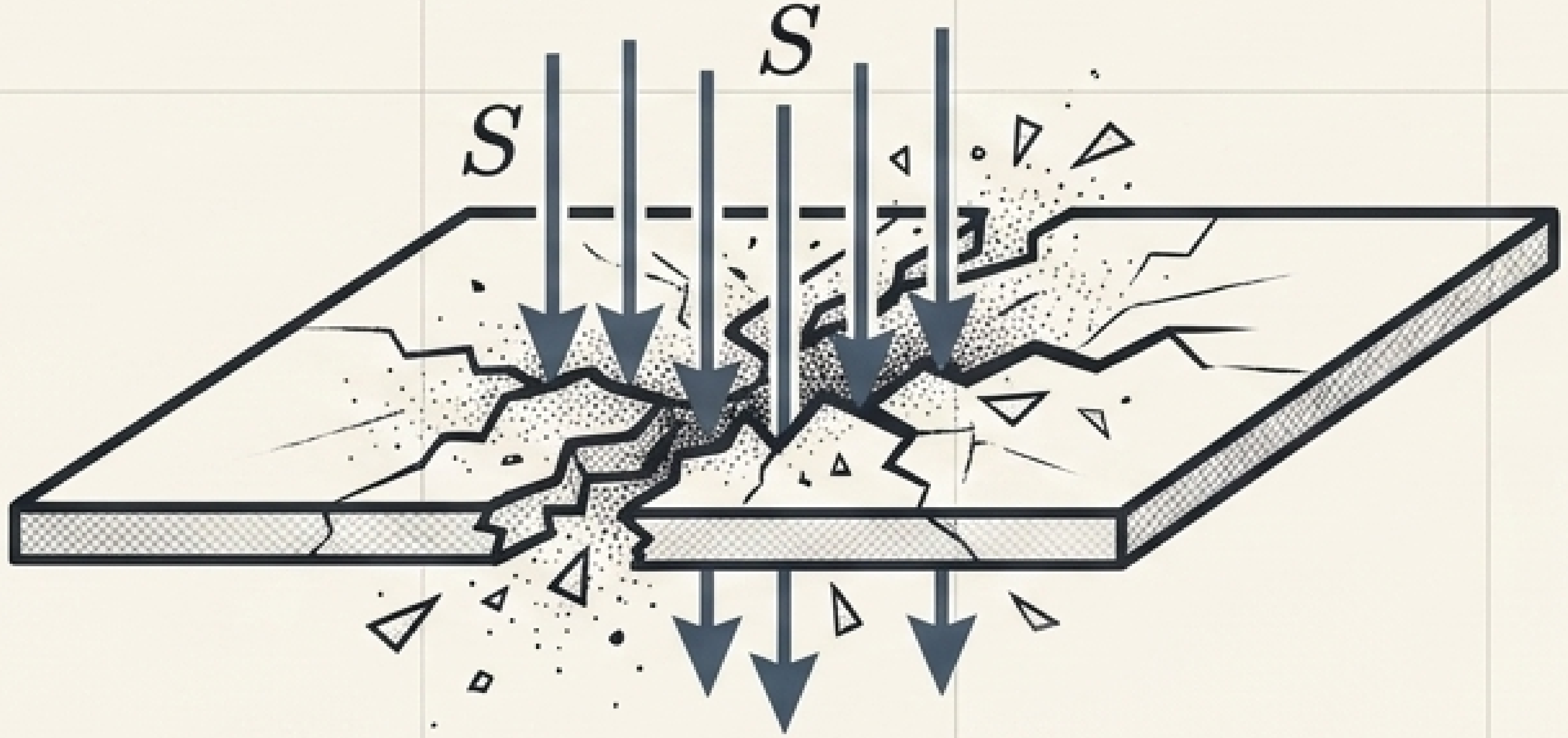
A system computes by maintaining a dynamical ensemble of trajectories whose time-averaged projection remains concentrated on M.

Function is not a property of a single trajectory; it is the projection of an ensemble.

Entropy Flow and Manifold Stability

Tangent Flow (ARE System)

Entropy as Resource: Entropy flow enlarges fibers and deepens cryptographic robustness while preserving the manifold.



Transverse Flow (Bacterial Uncoupling)

Entropy as Threat: Entropy crashes through the plane, destabilizing coupling and driving the trajectory out of the admissible regime.

Persistence ≠ low entropy.	Persistence = stability of invariant structure under entropy flow.

The Collapse of Microscopic Ontology

Classical ontology assumes the microscopic is foundational. But if stable function depends on equivalence classes and geometric projection, the exact microscopic realization is structurally secondary. Stability belongs not to the microscopic realization, but to the projection class itself.

The cryptography evaluator doesn't need the original messages.

The bacterial cell doesn't preserve transcription histories.

The pipeline breaks. The invariant holds.